Fiscal Year:	FY 2011	**Task Last Updated:**	FY 03/25/2011
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Fractional gravity studies of sensory mechanisms in root phototropism–a reflight experiment (TROPI-2)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jzk@fit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	336-334-5241
Organization Name:	University of North Carolina-Greensboro		
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City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	04/01/2009	**End Date:**	03/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
Contact Monitor:	Austin, Edward	**Contact Phone:**	650-604-2108
Contact Email:	edward.m.austin@nasa.gov		
Flight Program:	Shuttle/ISS		
Flight Assignment:	ISS STS-130 NOTE: New end date is 3/31/2012 per NSSC info (Ed., 2/19/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard (Miami University) Correll, Melanie (University of Florida)		
Grant/Contract No.:	NNX09AF11G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The major goals of this research are (1) to better understand cellular mechanisms of phototropism in plants and (2) to determine the effects and influence of gravity on light perception in plants. Specifically, we plan to use microgravity to facilitate studies of phototropism in roots and hypocotyls of young seedlings of the plant Arabidopsis thaliana. Results of our ground-based studies suggest that there are two photosensory systems that elicit phototropic responses in roots: a previously identified blue-light photoreceptor system and a red light phytochrome-based mechanism that we have recently discovered. Because of the interfering effect of the strong gravitropic response in roots, microgravity conditions are needed to effectively study this novel red-light-dependent phototropic response. Intermediate or fractional gravity studies (less than the nominal 1g on Earth) were planned to further study the interaction between tropisms. TROPI-1 was successfully performed on the ISS during late 2006. We obtained data on seedlings grown in microgravity and from the 1g control. However, one problem encountered during TROPI-1 was low seed germination due to long storage periods (8 months) in flight hardware. Thus, the originally proposed fractional gravity studies were not performed. TROPI-2 provides an opportunity to regain the results from these important fractional gravity experiments. These experiments will provide a better understanding of how plants integrate sensory input from multiple light and gravity perception systems. This information is important for growing plants on long-term space missions as part of life support systems. The fractional gravity studies contain 0.16g (Moon) and 0.38g (Mars) treatments, so information to be obtained is relevant to the exploration agenda at NASA. We used the European Modular Cultivation System (EMCS) on the ISS during Increment 22 in February/March 2010. Data analyses are still in progress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	From a basic science perspective, we will gain new insight into cellular/molecular events in light sensing. In the long term, through greater understanding of how plants perceive and respond to light, it will be possible to improve crop productivity. It also should be possible to modify agricultural crop plants with genetically improved phytochrome or their downstream signaling molecules to develop transgenic crop species with more fine tuned light-responsive developmental traits to better fit their ambient light environments.
Task Progress:	-Experiments were performed on the EMCS which is in the Columbus module of the ISS during Increment 22. -Two 6-day runs were performed. -The project was launched on Space Shuttle mission STS-130 and returned to Earth on STS-131. -Data retrieved during the experiments included video downlinks (from the ISS) of seedling growth, development, and tropisms. -Data retrieved after the experiment included samples frozen at -80C in MELFI. -Data analyses in progress include: image analysis and microarray experiments.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Kiss JZ, Correll MJ, Edelmann RE, Millar KDL. "Fractional gravity studies on the ISS of sensory mechanisms involved in phototropism." Committee on Space Research (COSPAR) 2010 38th Scientific Assembly, Bremen, Germany, July 18-25, 2010. COSPAR Abstract Book. Committee on Space Research (COSPAR) 2010 38th Scientific Assembly, Bremen, Germany, July 18-25, 2010. https://www.cospar-assembly.org/abstractcd/COSPAR-10/abstracts/data/pdf/abstracts/F12-0007-10.pdf , Jul-2010
Abstracts for Journals and Proceedings	Kiss JZ, Millar KDL, Kumar P, Correll MJ, Mullen JL, Hangarter RP, Edelmann RE. "Hypocotyls exhibit a phototropic response to red light in microgravity." Plant Biology 2010 (American Society of Plant Biologists annual meeting), Montreal, Canada, July 31-August 4, 2010. Abstract Book. Plant Biology 2010 (American Society of Plant Biologists annual meeting), 2010. http://abstracts.aspb.org/pb2010/public/M24/M2401.html , Aug-2010
Abstracts for Journals and Proceedings	Kiss JZ. "The use of the microgravity environment on the ISS as a research tool to study the mechanisms of tropisms (keynote speaker)." Keynote Speaker at the 24th Annual Meeting of the Japanese Society for Biological Sciences in Space, Sendai, Japan, September 17-18, 2010. Abstract Book, 24th Annual Meeting of the Japanese Society for Biological Sciences in Space, Sendai, Japan, September 17-18, 2010. p. 18. , Sep-2010
Abstracts for Journals and Proceedings	Huntoon KE, Millar KDL, Kiss JZ. "Phototropism is enhanced in seedlings grown in microgravity on the ISS." 26th Annual Meeting of the American Society for Gravitational and Space Biology, Washington, DC, November 4-7, 2010. Program and Abstracts, 26th Annual Meeting of the American Society for Gravitational and Space Biology, 2010. p. 35. , Nov-2010
Abstracts for Journals and Proceedings	Hall JS, Edelmann RE, Millar KDL, Kiss JZ. "TROPI-2: A study of phototropism in seedlings grown in fractional and microgravity on the ISS." 26th Annual Meeting of the American Society for Gravitational and Space Biology, Washington, DC, November 4-7, 2010. Program and Abstracts, 26th Annual Meeting of the American Society for Gravitational and Space Biology, 2010. p. 34. , Nov-2010
Abstracts for Journals and Proceedings	Pyle T, Millar KDL, Kumar P, Kiss JZ, Souret F, Correll MJ. "The effects of gravity and the spaceflight environment on the gene expression profile of Arabidopsis thaliana seedlings." 26th Annual Meeting of the American Society for Gravitational and Space Biology, Washington, DC, November 4-7, 2010. Program and Abstracts, 26th Annual Meeting of the American Society for Gravitational and Space Biology, 2010. p. 35. , Nov-2010

Articles in Peer-reviewed Journals	Kiss JZ, Millar KD, Kumar P, Edelmann RE, Correll MJ. "Improvements in the re-flight of spaceflight experiments on plant tropisms." Advances in Space Research. 2011 Feb 1;47(3):545-52. http://dx.doi.org/10.1016/j.asr.2010.09.024 , Feb-2011
Articles in Peer-reviewed Journals	Millar KD, Kumar P, Correll MJ, Mullen JL, Hangarter RP, Edelmann RE, Kiss JZ. "A novel phototropic response to red light is revealed in microgravity." New Phytologist 2010 May;186(3):648-56. Epub 2010 Mar 8. PMID: 20298479 , May-2010
Articles in Peer-reviewed Journals	Matía I, González-Camacho F, Herranz R, Kiss JZ, Gasset G, van Loon JJ, Marco R, Javier Medina F. "Plant cell proliferation and growth are altered by microgravity conditions in spaceflight." Journal of Plant Physiology 2010 Feb 15;167(3):184-93. Epub 2009 Oct 27. PMID: 19864040 ; http://dx.doi.org/10.1016/j.jplph.2009.08.012 , Feb-2010
Awards	Kiss JZ. "Orr E. Reynolds Distinguished Service Award, American Society for Gravitational and Space Biology, November 2010. This award is presented to a scientist in recognition for distinguished service to the Society 'above and beyond the call of duty.' " Nov-2010
Awards	Kiss JZ. "NASA Honor Award, 'for excellence in the category of a team (serving as PI) for the TROPI-2 experiment on the International Space Station. National Aeronautics and Space Administration, Ames Research Center,' May 2010." May-2010
Significant Media Coverage	Kiss JZ. "Interview on TROPI-2 ISS experiments. Univision network." Univision Radio, April 2010., Apr-2010
Significant Media Coverage	Kiss JZ. "Interview on TROPI-2 ISS experiments. Columbia National Radio Network." Columbia National Radio Network, February 2010., Feb-2010